

No	Particular	Theory		Practical	
		Amplitude	frequency	Amplitude	Time Frequency
1.	message signal	1V	70Hz	1V	$T_m = x_2 - x_1$ $= 0.0143s$ $f_m = 70Hz$
2.	Carrier signal	3V	700Hz	3V	$T_c = x_2 - x_1$ $= 0.0143s$ $f_c = 700Hz$
3.	AM signal (DSB-SC)	-	-	A_{DSB-SC} $= A_m \times A_c$ $= 3V$	$T_m = 0.0143s$ $f_m = 700Hz$

Ex 2
22/07/26

DOUBLE SIDEBAND SUPPRESSED CARRIER

[DSB-SC]

Aim:

To generate the double side band suppressed carrier by using SCI LAB.

APPARATUS REQUIRED

* Computer with i3 processor

* SCI LAB software.

CODE:

```
clc;  
clear;  
clf;  
am = 1;  
ac = 3;  
fm = 70;  
fc = 700;  
ts = 10000;  
t = 0:1/ts:0.1;  
m = am * cos(2 * %pi * fm * t);  
subplot(3,1,1);  
plot(t,m);
```

Calculation:

1. Message Signal

Practical time:-

$$T_m = x_2 - x_1$$

$$= -0.025 + 0.035$$

$$= 0.0143$$

Practical

Frequency

$$f_m = \frac{1}{T_m}$$

$$= \frac{1}{0.0143}$$

2. Carrier Signal

Practical Time

$$T_c = x_2 - x_1$$

$$= -0.0015 + 0.00293$$

$$= 0.00143 \text{ s}$$

Frequency:-

$$F_c = \frac{1}{T_c}$$

$$= 700 \text{ Hz}$$

xtitle ("Message Signal");

c = ac * cos (2 * %pi * tc * t);

subplot (3,1,2);

plot (t, c);

xtitle ("Carrier Signal");

dsbssc = m.*c;

subplot (3,1,3);

plot (t, dsbssc);

xtitle ("DSB-SC Signal");

3. DSB-SC (Signal "Message") title x

Practical Amplitude

$$A_{DSB} = A_m \times A_c$$

$$= 1 \times 3$$

$$= 3V$$

Practical Time

$$T_{DSB} = X_2 - X_1$$

$$= 0.00284 - 0.00141$$

$$T_{DSB} = 0.00143s$$

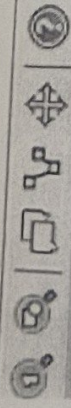
Practical frequency :-

$$f_{DSB} = \frac{1}{T_{DSB}}$$

$$= 700 \text{ Hz}$$

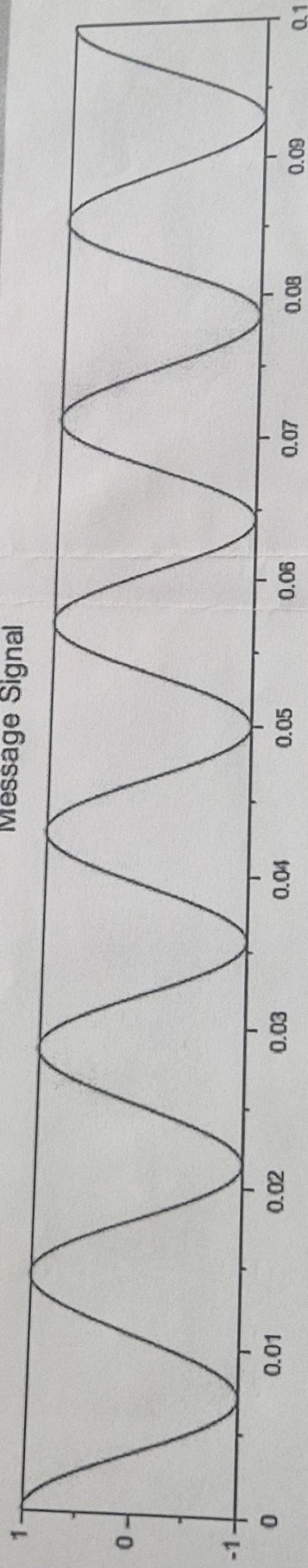
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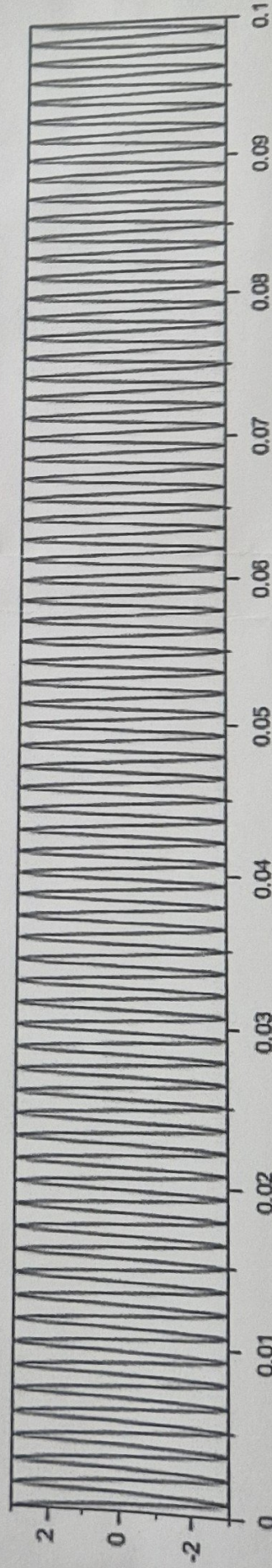


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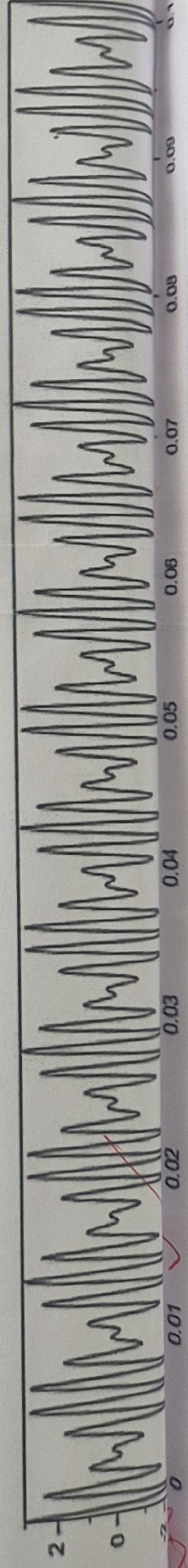
Message Signal



Carrier Signal



DSB-SC Signal



Sno	Content	Mark	Mark obtained
1.	Program implemtation	20	20
2.	wave form and tabulation	20	20
3.	Viva voce	20	15
4.	Calculation	20	20
5.	Result / Observation	20	20
Total			95

Result: -

Thus the double side band Suppressed
Carrier (DSB-SC) successfully run on the
SCI LAB